

Laboratorio 6

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Provision an Azure SQL Database

Create Server

Home > Azure SQL | SQL databases > Create SQL Database

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name *

server-lab6-alumno09

.database.windows.net

Location *

(Europe) Spain Central

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#)

Authentication method

☐ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☒ Use SQL authentication

Server admin login *

adminlab6

Password *

.....

Confirm password *

.....

OK

Contraseña: @Dmin123

Create Database Basic

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

✖ Basics

Networking

Security

Additional settings

Tags

Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Create a serverless Azure SQL Database with 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of your subscription. Limit is 10 free databases per subscription. [Learn more](#)

Apply offer

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Azure for Students

Resource group * ⓘ

Select a resource group

Create new

Database details

Enter required settings for this database, including server name, database name, and storage.

Database name *

Server * ⓘ

Review + create

Next : Networking >

A resource group is a container that holds related resources for an Azure solution.

Name *

resourcegroup-lab6-alumno09

OK

Cancel

Home > Azure SQL | SQL databases

Create SQL Database ...

Microsoft

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

AdventureWorksLT

Server * ⓘ

(new) server-lab6-alumno09 (Spain Central)

Create new

Want to use SQL elastic pool? ⓘ

☐ Yes

☒ No

Workload environment

☒ Development

☐ Production

Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * ⓘ

Hyperscale - Serverless

Standard-series (Gen5), 2 vCores

Configure database

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

The default backup storage redundancy setting is taken from the setting of the source.

Backup storage redundancy ⓘ

☒ Locally-redundant backup storage

☐ Zone-redundant backup storage

Compute + Storage

Home > Azure SQL | SQL databases > Create SQL Database

Configure ...

 Feedback

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

 SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier

Hyperscale (Highly scalable compute and storage)

[Compare service tiers](#)

 [Learn more](#) about migrating your data into Hyperscale.

Compute tier

☐

Provisioned - Compute resources are pre-allocated. Billed per hour based on vCores configured.

☒

Serverless - Compute resources are auto-scaled. Billed per second based on vCores used.

Create Database Networking

Home > Azure SQL | SQL databases

Create SQL Database ...

Microsoft

Basics **Networking** Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'server-lab6-alumno09' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

- Connectivity method * ⓘ
- ☐ No access
 - ☒ Public endpoint
 - ☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

Connection policy

Configure how clients communicate with your SQL database server. [Learn more](#)

- Connection policy ⓘ
- ☒ Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure
 - ☐ Proxy - All connections are proxied via the Azure SQL Database gateways
 - ☐ Redirect - Clients establish connections directly to the node hosting the database

Create Database Additional Settings

Home > Azure SQL | SQL databases

Create SQL Database ...

Microsoft

Basics Networking Security Additional settings Tags Review + create

Customize additional configuration parameters including collation & sample data.

Data source

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data * None Backup Sample

AdventureWorksLT will be created as the sample database.

Database collation

Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL_Latin1_General_CP1_CI_AS. [Learn more](#)

Collation ⓘ SQL_Latin1_General_CP1_CI_AS

Review + create < Previous Next : Tags >

Create the test workload

Conectar BD en SSMS

Conectar

History Examinar

Escriba aquí para filtrar la lista

▼ Favoritos
No hay conexiones favoritas

▼ Conexiones recientes (3)

- server-lab6-alumno09.database.windows.net, <predeterminado> ...
- localhost, <predeterminado> (sa)
- t07w02, <predeterminado> (sa)

Propiedades de conexión Cadena de conexión

Nombre del servidor server-lab6-alumno09.database.windows.net ▼

Authentication Autenticación de SQL Server ▼

Nombre de usuario adminlab6

Contraseña

☒ Recordar contraseña

Nombre de la base de datos <predeterminado> ▼

Cifrar Obligatorio ▼

☒ Certificado de servidor de confianza

Nombre de host en el certificado

Avanzadas...

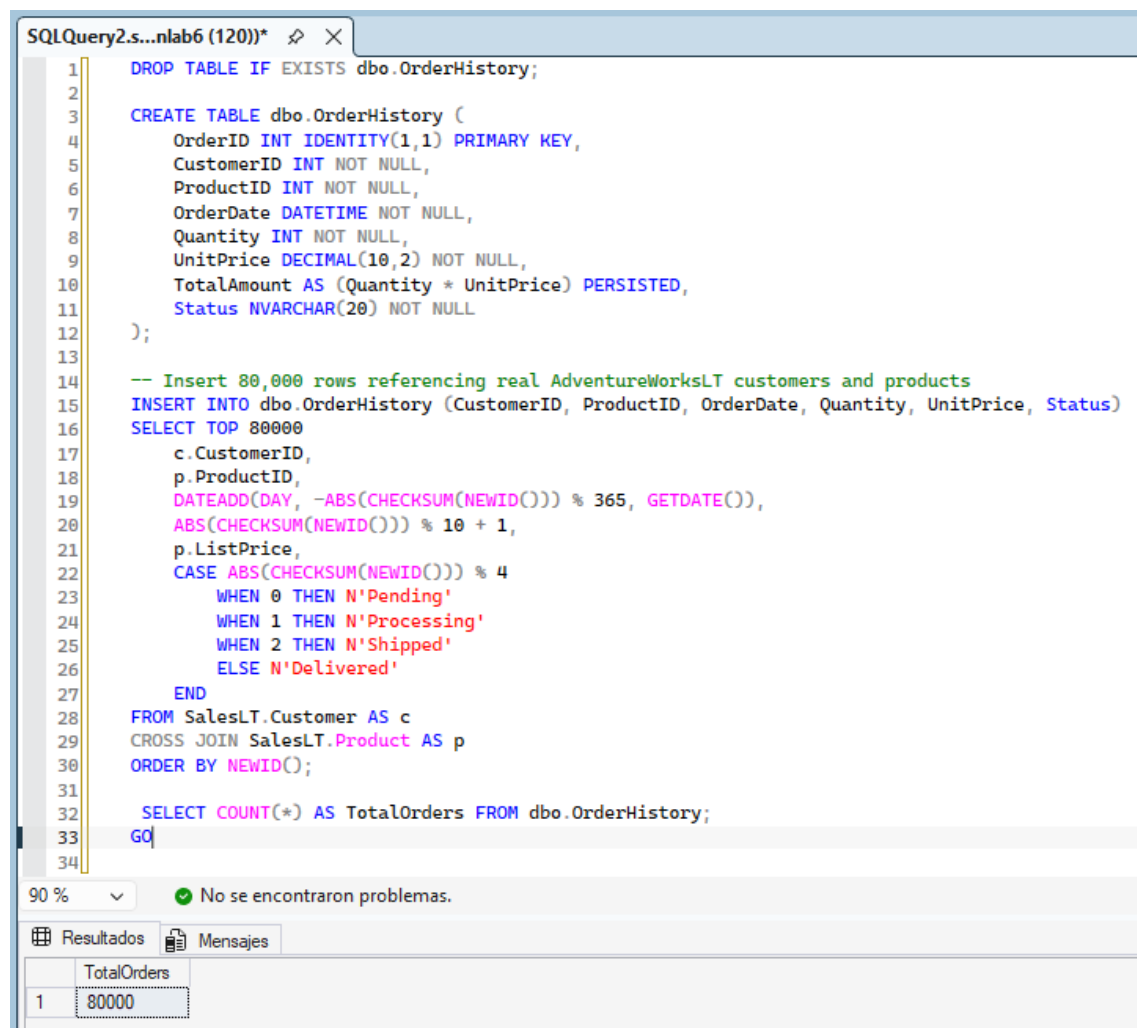
Propiedades personalizadas

Nombre

Color <predeterminado> ▼ Personalizado...

Restablecer Conectar Cancelar

Script Order History



SQLQuery2.s...nlab6 (120))*

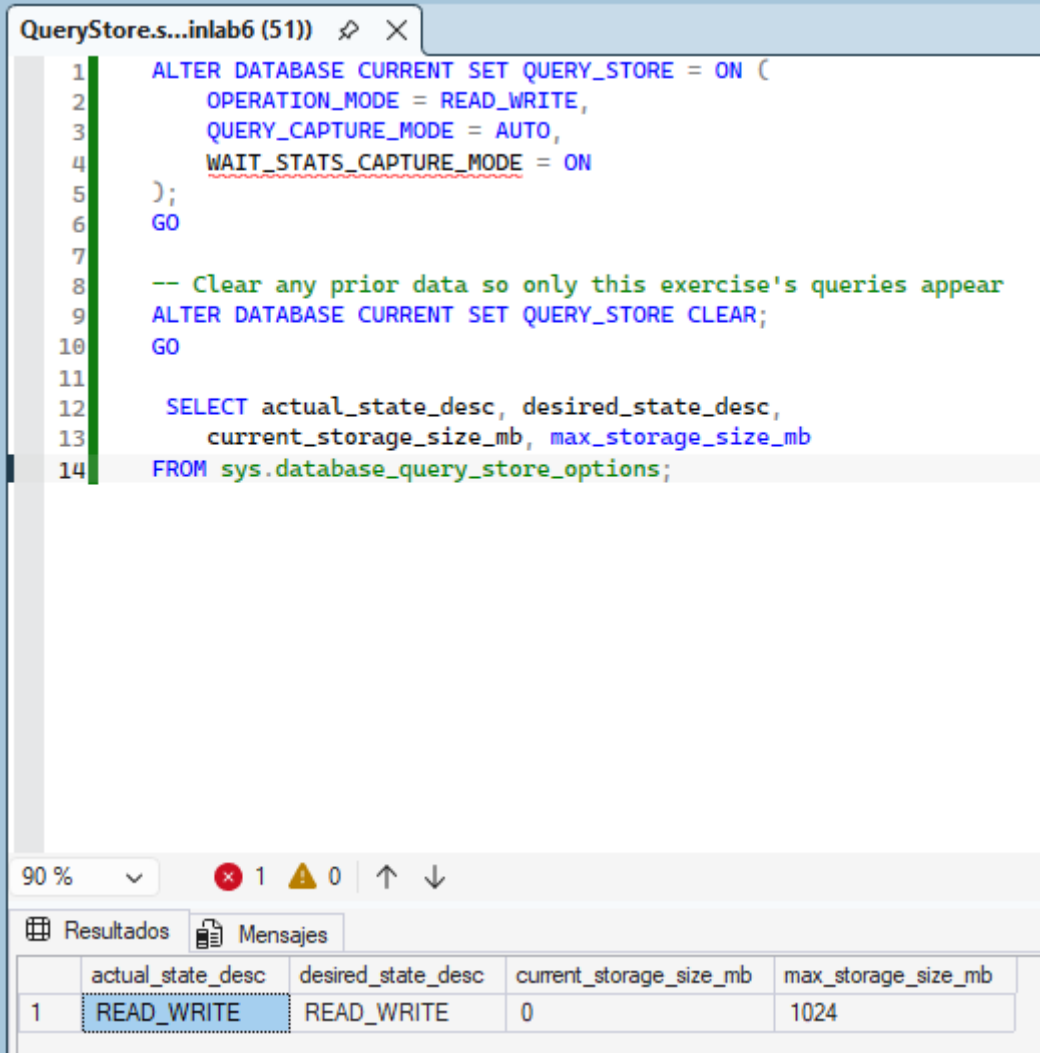
```
1 DROP TABLE IF EXISTS dbo.OrderHistory;
2
3 CREATE TABLE dbo.OrderHistory (
4     OrderID INT IDENTITY(1,1) PRIMARY KEY,
5     CustomerID INT NOT NULL,
6     ProductID INT NOT NULL,
7     OrderDate DATETIME NOT NULL,
8     Quantity INT NOT NULL,
9     UnitPrice DECIMAL(10,2) NOT NULL,
10    TotalAmount AS (Quantity * UnitPrice) PERSISTED,
11    Status NVARCHAR(20) NOT NULL
12 );
13
14 -- Insert 80,000 rows referencing real AdventureWorksLT customers and products
15 INSERT INTO dbo.OrderHistory (CustomerID, ProductID, OrderDate, Quantity, UnitPrice, Status)
16 SELECT TOP 80000
17     c.CustomerID,
18     p.ProductID,
19     DATEADD(DAY, -ABS(CHECKSUM(NEWID())) % 365, GETDATE()),
20     ABS(CHECKSUM(NEWID())) % 10 + 1,
21     p.ListPrice,
22     CASE ABS(CHECKSUM(NEWID())) % 4
23         WHEN 0 THEN N'Pending'
24         WHEN 1 THEN N'Processing'
25         WHEN 2 THEN N'Shipped'
26         ELSE N'Delivered'
27     END
28 FROM SalesLT.Customer AS c
29 CROSS JOIN SalesLT.Product AS p
30 ORDER BY NEWID();
31
32 SELECT COUNT(*) AS TotalOrders FROM dbo.OrderHistory;
33 GO
34
```

90 % No se encontraron problemas.

Resultados Mensajes

	TotalOrders
1	80000

Enable and configure Query Store



The screenshot shows a SQL Server Enterprise Manager interface. The top pane displays a query window titled "QueryStore.s...inlab6 (51)". The query contains the following T-SQL code:

```
1 ALTER DATABASE CURRENT SET QUERY_STORE = ON (
2     OPERATION_MODE = READ_WRITE,
3     QUERY_CAPTURE_MODE = AUTO,
4     WAIT_STATS_CAPTURE_MODE = ON
5 );
6 GO
7
8 -- Clear any prior data so only this exercise's queries appear
9 ALTER DATABASE CURRENT SET QUERY_STORE CLEAR;
10 GO
11
12 SELECT actual_state_desc, desired_state_desc,
13        current_storage_size_mb, max_storage_size_mb
14 FROM sys.database_query_store_options;
```

Below the query window, the "Resultados" (Results) tab is active, showing a grid with the following data:

	actual_state_desc	desired_state_desc	current_storage_size_mb	max_storage_size_mb
1	READ_WRITE	READ_WRITE	0	1024

Analyze execution plans and add a missing index

Actual Execution Plan

 Formato de SQL (versión preliminar)	Ctrl+K, Ctrl+Q
 Cortar	Ctrl+X
 Copiar	Ctrl+C
 Pegar	Ctrl+V
 Insertar fragmento de código...	Ctrl+K, Ctrl+X
 Rodear con...	Ctrl+K, Ctrl+S
Conexión	>
Abrir servidor en Explorador de objetos	Alt+F8
 Ejecutar	Alt+X
 Mostrar plan de ejecución estimado	Ctrl+L
 IntelliSense habilitado	Ctrl+B, Ctrl+I
 Realizar seguimiento de la consulta en SQL Server Profiler	Ctrl+Alt+P
 Analizar la consulta en el Asistente para la optimización del motor de base de datos	
 Diseñar consulta en el editor...	Ctrl+Mayús.+Q
 Incluir plan de ejecución actual	Ctrl+M
 Incluir estadísticas de consultas dinámicas	
 Incluir estadísticas de cliente	Mayús.+Alt+S
Resultados a	>
 Ventana Propiedades	F4
 Opciones de Query...	

Statistics Query

SQLQuery4.s...inlab6 (51))*

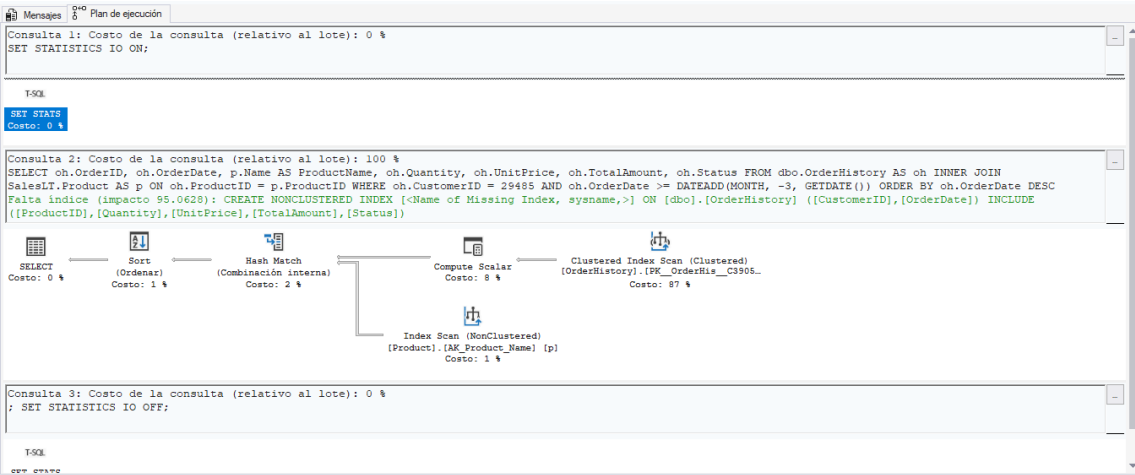
```
1 SET STATISTICS IO ON;
2 SELECT
3     oh.OrderID,
4     oh.OrderDate,
5     p.Name AS ProductName,
6     oh.Quantity,
7     oh.UnitPrice,
8     oh.TotalAmount,
9     oh.Status
10 FROM dbo.OrderHistory AS oh
11 INNER JOIN SalesLT.Product AS p
12     ON oh.ProductID = p.ProductID
13 WHERE oh.CustomerID = 29485
14     AND oh.OrderDate >= DATEADD(MONTH, -3, GETDATE())
15 ORDER BY oh.OrderDate DESC;
16 SET STATISTICS IO OFF;
```

82 % 11 0

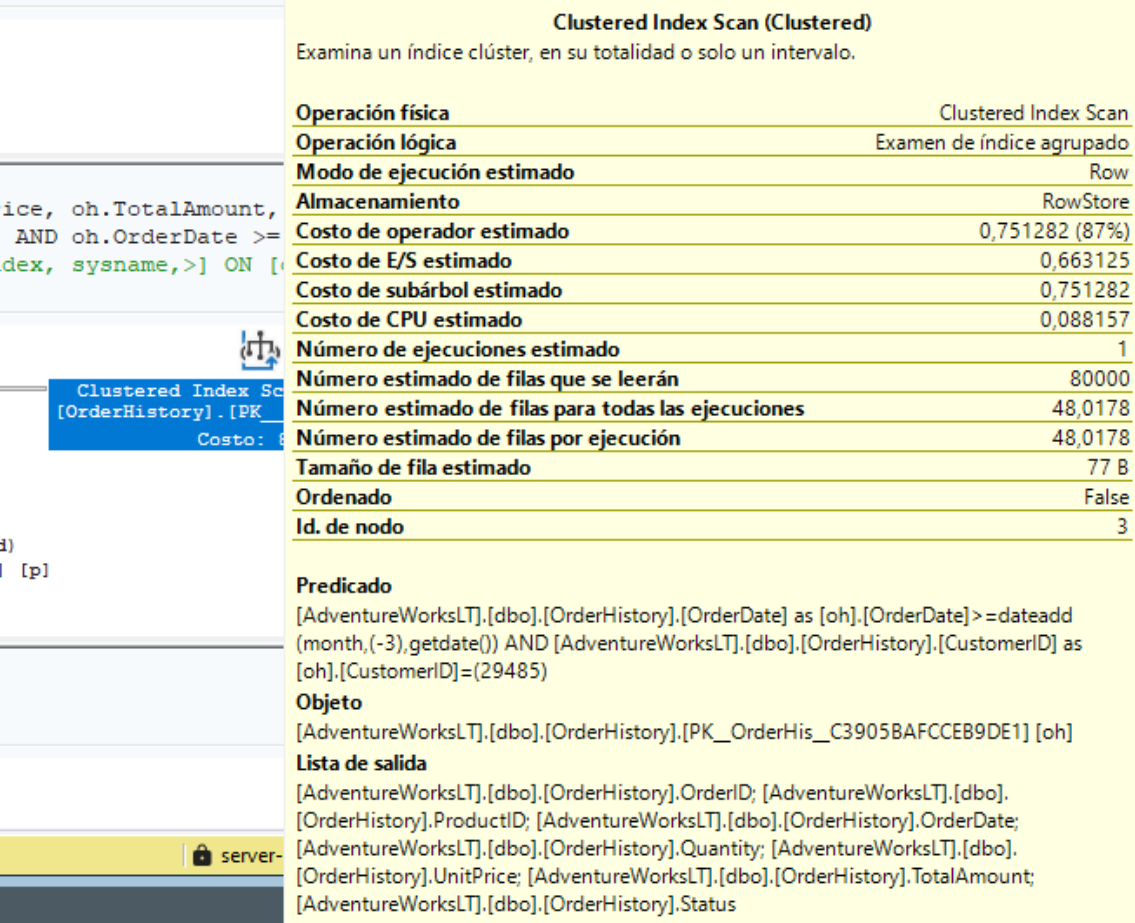
Resultados Mensajes

	OrderID	OrderDate	ProductName	Quantity	UnitPrice	TotalAmount	Status
1	2366	2026-10-06 11:24:46.293	Sport-100 Helmet, Red	7	34.99	244.93	Delivered
2	76445	2026-09-29 11:24:46.293	HL Road Rear Wheel	2	357.06	714.12	Pending
3	36011	2026-09-28 11:24:46.293	Classic Vest, S	6	63.50	381.00	Shipped
4	26627	2026-09-22 11:24:46.293	Touring Tire Tube	6	4.99	29.94	Pending
5	8131	2026-09-20 11:24:46.293	Full-Finger Gloves, M	1	37.99	37.99	Delivered
6	68734	2026-09-19 11:24:46.293	ML Road Rear Wheel	1	275.39	275.39	Processing
7	46407	2026-09-18 11:24:46.293	Mountain Bike Socks, M	3	9.50	28.50	Delivered
8	55860	2026-09-07 11:24:46.293	Mountain-500 Black, 52	3	539.99	1619.97	Shipped
9	15325	2026-09-07 11:24:46.293	Long-Sleeve Logo Jersey, M	6	49.99	299.94	Delivered
10	25945	2026-09-07 11:24:46.293	Touring-1000 Yellow, 54	8	2384.07	19072.56	Pending
11	40958	2026-09-04 11:24:46.293	Full-Finger Gloves, L	10	37.99	379.90	Delivered
12	23385	2026-09-03 11:24:46.293	HL Road Tire	9	32.60	293.40	Delivered
13	38463	2026-08-18 11:24:46.293	Short-Sleeve Classic Jersey, M	7	53.99	377.93	Delivered
14	51674	2026-08-17 11:24:46.293	Short-Sleeve Classic Jersey, XL	8	53.99	431.92	Delivered
15	73851	2026-08-06 11:24:46.293	Road-350-W Yellow, 40	6	1700.99	10205.94	Processing
16	34746	2026-08-04 11:24:46.293	All-Purpose Bike Stand	6	159.00	954.00	Delivered
17	37479	2026-07-28 11:24:46.293	Road-250 Red, 58	10	2443.35	24433.50	Processing
18	67481	2026-07-27 11:24:46.293	ML Mountain Frame - Black, 38	6	348.76	2092.56	Pending
19	52204	2026-07-24 11:24:46.293	HL Mountain Frame - Black, 42	7	1349.60	9447.20	Delivered
20	17470	2026-07-22 11:24:46.293	HL Touring Seat/Saddle	2	52.64	105.28	Shipped
21	9180	2026-07-19 11:24:46.293	Touring-3000 Blue, 62	9	742.35	6681.15	Processing
22	22587	2026-07-18 11:24:46.293	Short-Sleeve Classic Jersey, L	6	53.99	323.94	Processing
23	28319	2026-07-18 11:24:46.293	HL Road Frame - Red, 56	2	1431.50	2863.00	Processing
24	47984	2026-07-15 11:24:46.293	Road Bottle Cage	1	8.99	8.99	Delivered
25	54093	2026-07-14 11:24:46.293	LL Mountain Frame - Black, 40	3	249.79	749.37	Delivered
26	16990	2026-07-13 11:24:46.293	Road-650 Black, 52	8	782.99	6263.92	Shipped
27	44966	2026-07-08 11:24:46.293	HL Headset	2	124.73	249.46	Delivered

Execution Plan



Clustered Index Scan OrderHistory



IndexScan RowNumbers



```
= Index Scan (NonClustered)
[Product].[AK_Product_Name]
Costo: 1 %
```

Costo de ejecución estimado	
Almacenamiento	RowStore
Costo de operador estimado	0,005088 (1%)
Costo de E/S estimado	0,0046065
Costo de subárbol estimado	0,005088
Costo de CPU estimado	0,0004815
Número de ejecuciones estimado	1
Número estimado de filas que se leerán	295
Número estimado de filas para todas las ejecuciones	295
Número estimado de filas por ejecución	295

Missing Index

Falta indice (impacto 95.0628): CREATE NONCLUSTERED INDEX [<Name of Missing Index, sysname,>] ON [dbo].[OrderHistory] (([CustomerID],[OrderDate])) INCLUDE (([ProductID],[Quantity],[UnitPrice],[TotalAmount],[Status]))

Index

```
Index.sql - ...inlab6 (118)
1  /*
2  Faltan detalles del indice de QueryStatistics.sql - server-lab6-alumno09.database.windows.net.AdventureWorksLT (adminlab6 (51))
3  El procesador de consultas estima que la implementación del siguiente indice podria mejorar el costo de la consulta en un 95.0628%.
4  */
5
6  /*
7  USE [AdventureWorksLT]
8  GO
9  CREATE NONCLUSTERED INDEX [<Name of Missing Index, sysname,>]
10 ON [dbo].[OrderHistory] (([CustomerID],[OrderDate]))
11 INCLUDE ([ProductID],[Quantity],[UnitPrice],[TotalAmount],[Status])
12 GO
13 */
14
```

Logical Reads

34

82 %  No se encontraron problemas.

 Resultados

 Mensajes

(80000 filas afectadas)

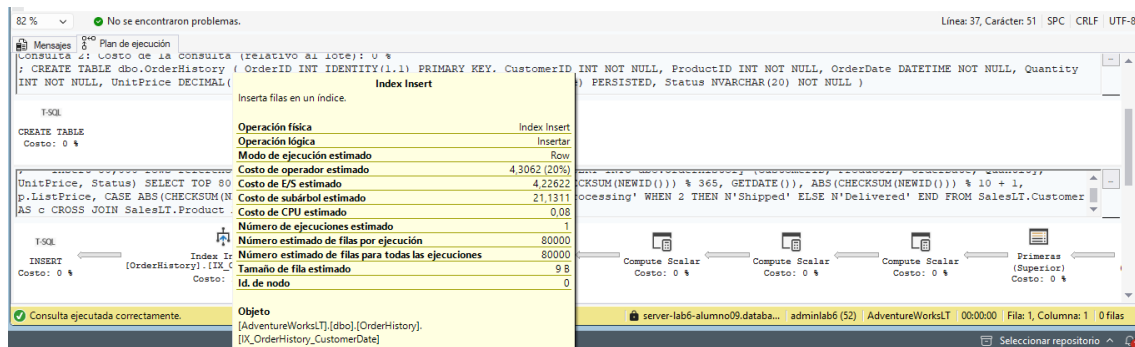
(1 fila afectada)

Hora de finalización: 2026-10-07T13:52:26.9053295+02:00

Create Cover Index

```
34
35 CREATE NONCLUSTERED INDEX IX_OrderHistory_CustomerDate
36 ON dbo.OrderHistory (CustomerID, OrderDate DESC)
37 INCLUDE (ProductID, Quantity, UnitPrice, Status);
```

Index Seek



82 % No se encontraron problemas. Línea: 37, Carácter: 51 SPC CRLF UTF-8

Consultas: Costo de la consulta (relativo al 1056): 0 %

CREATE TABLE dbo.OrderHistory (OrderID INT IDENTITY(1,1) PRIMARY KEY, CustomerID INT NOT NULL, ProductID INT NOT NULL, OrderDate DATETIME NOT NULL, Quantity INT NOT NULL, UnitPrice DECIMAL(10,2) NOT NULL, Status NVARCHAR(20) NOT NULL)

Inserta filas en un índice. Index Insert

Operación física Index Insert

Operación lógica Insertar

Modo de ejecución estimado Row

Costo de operador estimado 4,3062 (20%)

Costo de E/S estimado 4,22622

Costo de subárbol estimado 21,1311

Costo de CPU estimado 0,08

Número de ejecuciones estimado 1

Número estimado de filas por ejecución 80000

Número estimado de filas para todas las ejecuciones 80000

Tamaño de fila estimado 9,8

Id. de nodo 0

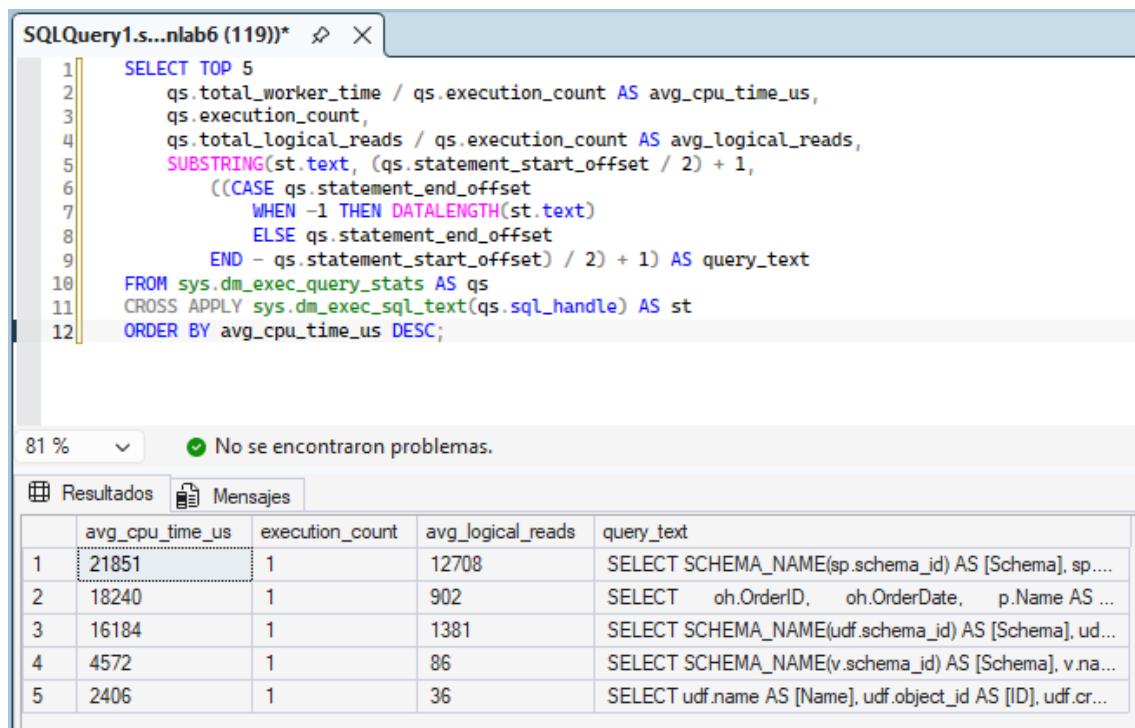
Objeto [AdventureWorksLT].[dbo].[OrderHistory].[IX_OrderHistory_CustomerDate]

Consulta ejecutada correctamente.

server-lab6-alumno09.databa... adminlab6 (52) AdventureWorksLT 00:00:00 Fila: 1, Columna: 1 0 filas

Use DMVs to find the most expensive queries

CPU Time



SQLQuery1.s...nlab6 (119))*

```
1 SELECT TOP 5
2     qs.total_worker_time / qs.execution_count AS avg_cpu_time_us,
3     qs.execution_count,
4     qs.total_logical_reads / qs.execution_count AS avg_logical_reads,
5     SUBSTRING(st.text, (qs.statement_start_offset / 2) + 1,
6         ((CASE qs.statement_end_offset
7             WHEN -1 THEN DATALength(st.text)
8             ELSE qs.statement_end_offset
9         END - qs.statement_start_offset) / 2) + 1) AS query_text
10 FROM sys.dm_exec_query_stats AS qs
11 CROSS APPLY sys.dm_exec_sql_text(qs.sql_handle) AS st
12 ORDER BY avg_cpu_time_us DESC;
```

81 % No se encontraron problemas.

Resultados Mensajes

	avg_cpu_time_us	execution_count	avg_logical_reads	query_text
1	21851	1	12708	SELECT SCHEMA_NAME(sp.schema_id) AS [Schema], sp....
2	18240	1	902	SELECT oh.OrderID, oh.OrderDate, p.Name AS ...
3	16184	1	1381	SELECT SCHEMA_NAME(udf.schema_id) AS [Schema], udf...
4	4572	1	86	SELECT SCHEMA_NAME(v.schema_id) AS [Schema], v.na...
5	2406	1	36	SELECT udf.name AS [Name], udf.object_id AS [ID], udf.cr...

5 Tablas

SQLQuery2.s...nlab6 (122))* Top5_CPUTime...inlab6 (119))

```

1  SELECT
2      p.Name AS ProductName,
3      p.ProductNumber,
4      oh.OrderDate,
5      oh.Quantity,
6      oh.TotalAmount
7  FROM dbo.OrderHistory AS oh
8  INNER JOIN SalesLT.Product AS p
9      ON oh.ProductID = p.ProductID
10 WHERE oh.Status = N'Pending'
11 AND oh.OrderDate >= DATEADD(MONTH, -1, GETDATE())
12 ORDER BY oh.TotalAmount DESC;
13 GO 5

```

81 % 2 0 ↑ ↓

Resultados Mensajes

	ProductName	ProductNumber	OrderDate	Quantity	TotalAmount
1	Road-150 Red, 62	BK-R93R-62	2026-09-27 12:14:27.227	10	35782.70
2	Road-150 Red, 56	BK-R93R-56	2026-09-18 12:14:27.227	10	35782.70
3	Road-150 Red, 56	BK-R93R-56	2026-09-30 12:14:27.227	10	35782.70
4	Mountain-100 Silver, 38	BK-M82S-38	2026-09-28 12:14:27.227	10	33999.90

	ProductName	ProductNumber	OrderDate	Quantity	TotalAmount
1	Road-150 Red, 62	BK-R93R-62	2026-09-27 12:14:27.227	10	35782.70
2	Road-150 Red, 56	BK-R93R-56	2026-09-18 12:14:27.227	10	35782.70
3	Road-150 Red, 56	BK-R93R-56	2026-09-30 12:14:27.227	10	35782.70
4	Mountain-100 Silver, 38	BK-M82S-38	2026-09-28 12:14:27.227	10	33999.90

	ProductName	ProductNumber	OrderDate	Quantity	TotalAmount
1	Road-150 Red, 62	BK-R93R-62	2026-09-27 12:14:27.227	10	35782.70
2	Road-150 Red, 56	BK-R93R-56	2026-09-18 12:14:27.227	10	35782.70
3	Road-150 Red, 56	BK-R93R-56	2026-09-30 12:14:27.227	10	35782.70
4	Mountain-100 Silver, 38	BK-M82S-38	2026-09-28 12:14:27.227	10	33999.90

	ProductName	ProductNumber	OrderDate	Quantity	TotalAmount
1	Road-150 Red, 62	BK-R93R-62	2026-09-27 12:14:27.227	10	35782.70
2	Road-150 Red, 56	BK-R93R-56	2026-09-18 12:14:27.227	10	35782.70
3	Road-150 Red, 56	BK-R93R-56	2026-09-30 12:14:27.227	10	35782.70
4	Mountain-100 Sil...	BK-M82S-38	2026-09-28 12:14:27.227	10	33999.90

	ProductName	ProductNumber	OrderDate	Quantity	TotalAmount
1	Road-150 Red, 62	BK-R93R-62	2026-09-27 12:14:27.227	10	35782.70
2	Road-150 Red, 56	BK-R93R-56	2026-09-18 12:14:27.227	10	35782.70
3	Road-150 Red, 56	BK-R93R-56	2026-09-30 12:14:27.227	10	35782.70
4	Mountain-100 Sil...	BK-M82S-38	2026-09-28 12:14:27.227	10	33999.90
5	Mountain-100 Sil...	BK-M82S-42	2026-09-12 12:14:27.227	10	33999.90
6	Mountain-100 Sil...	BK-M82S-44	2026-09-19 12:14:27.227	10	33999.90
7	Mountain-100 Sil...	BK-M82S-44	2026-10-07 12:14:27.227	10	33999.90
8	Mountain-100 Bl...	BK-M82B-48	2026-09-08 12:14:27.227	10	33749.90
9	Mountain-100 Bl...	BK-M82B-44	2026-09-10 12:14:27.227	10	33749.90

✓ Consulta ejecutada correctamente.

Optimizer Recommendation

SQLQuery3.s...inlab6 (51)) 5Tablas.sql -...minlab6 (122) Top5_CPUTime...inlab6 (119))

```
1 SELECT TOP 10
2     mid.statement AS table_name,
3     mid.equality_columns,
4     mid.inequality_columns,
5     mid.included_columns,
6     ROUND(migs.avg_total_user_cost * migs.avg_user_impact *
7           (migs.user_seeks + migs.user_scans), 2) AS improvement_measure
8 FROM sys.dm_db_missing_index_groups AS mig
9 INNER JOIN sys.dm_db_missing_index_group_stats AS migs
10 ON migs.group_handle = mig.index_group_handle
11 INNER JOIN sys.dm_db_missing_index_details AS mid
12 ON mig.index_handle = mid.index_handle
13 ORDER BY improvement_measure DESC;
```

81 % No se encontraron problemas.

Resultados Mensajes

	table_name	equality_columns	inequality_columns	included_columns	improvement_measure
1	[AdventureWorksLT].[dbo].[OrderHistory]	[ProductID], [Status]	[OrderDate]	[Quantity], [TotalAmount]	3264,86
2	[AdventureWorksLT].[dbo].[OrderHistory]	[Status]	[OrderDate]	[ProductID], [Quantity], [TotalAmount]	2419,75

Simulate and detect a plan regression with Query

Set up regression

```
SQLQuery1.s...lab6 (115))* ✕ X
1  ALTER DATABASE CURRENT SET QUERY_STORE CLEAR;
2  GO
3  ALTER DATABASE CURRENT SET QUERY_STORE (QUERY_CAPTURE_MODE = ALL);
4  GO
5  DROP INDEX IF EXISTS IX_OrderHistory_CustomerDate ON dbo.OrderHistory;
6  CREATE NONCLUSTERED INDEX IX_OrderHistory_CustomerDate
7  ON dbo.OrderHistory (CustomerID, OrderDate DESC);
8  GO
9  INSERT INTO dbo.OrderHistory (CustomerID, ProductID, OrderDate, Quantity, UnitPrice, Status)
10 SELECT TOP 50000
11     1,
12     p.ProductID,
13     DATEADD(DAY, -ABS(CHECKSUM(NEWID())) % 365, GETDATE()),
14     ABS(CHECKSUM(NEWID())) % 10 + 1,
15     p.ListPrice,
16     N'Pending'
17 FROM SalesLT.Product AS p
18 CROSS JOIN SalesLT.Product AS p2
19 ORDER BY NEWID();
20 GO
21 UPDATE STATISTICS dbo.OrderHistory;
22 GO
23 CREATE OR ALTER PROCEDURE dbo.GetCustomerOrders
24     @CustomerID INT
25 AS
26 BEGIN
27     SELECT
28         oh.OrderID,
29         oh.OrderDate,
30         p.Name AS ProductName,
31         oh.Quantity,
32         oh.UnitPrice,
33         oh.TotalAmount,
34         oh.Status
35 FROM dbo.OrderHistory AS oh
36 INNER JOIN SalesLT.Product AS p
37     ON oh.ProductID = p.ProductID
38 WHERE oh.CustomerID = @CustomerID
39 ORDER BY oh.OrderDate DESC;
40 END;
```

75 % ✕ 1 ⚠ 0 ↑ ↓

Mensajes

(50000 filas afectadas)

Hora de finalización: 2026-10-07T19:35:05.1318291+02:00

Cause Regression

SQLQuery2.s...lab6 (132)*

```
1 ALTER DATABASE SCOPED CONFIGURATION CLEAR PROCEDURE_CACHE;
2 GO
3 EXEC dbo.GetCustomerOrders @CustomerID = 29485;
4 GO 10
5 ALTER DATABASE SCOPED CONFIGURATION CLEAR PROCEDURE_CACHE;
6 GO
7 EXEC dbo.GetCustomerOrders @CustomerID = 1;
8 GO
9 EXEC dbo.GetCustomerOrders @CustomerID = 29485;
10 GO 10
11 EXEC sp_query_store_flush_db;
```

110 % 7 0

Linea: 11, Carácter: 2 TABULACIONES CRLF UTF-8

Resultados Mensajes

OrderID	OrderDate	ProductName	Quantity	UnitPrice	TotalAmount	Status
15096	2026-10-07 12:14:27.227	HL Road Frame - Black, 44	7	1431.50	10020.50	Delivered
38966	2026-10-03 12:14:27.227	Touring-1000 Yellow, 60	1	2384.07	2384.07	Delivered
24456	2026-10-02 12:14:27.227	Racing Socks, L	4	8.99	35.96	Delivered
9463	2026-09-23 12:14:27.227	Men's Bib-Shorts, S	8	89.99	719.92	Shipped
69914	2026-09-22 12:14:27.227	Front Brakes	10	106.50	1065.00	Pending
30858	2026-09-15 12:14:27.227	ML Road Tire	9	24.99	224.91	Shipped
41047	2026-09-15 12:14:27.227	Road-650 Red, 58	2	782.99	1565.98	Pending
20941	2026-09-12 12:14:27.227	Road-250 Red, 44	8	2443.35	19546.80	Delivered

OrderID	OrderDate	ProductName	Quantity	UnitPrice	TotalAmount	Status
15096	2026-10-07 12:14:27.227	HL Road Frame - Black, 44	7	1431.50	10020.50	Delivered
38966	2026-10-03 12:14:27.227	Touring-1000 Yellow, 60	1	2384.07	2384.07	Delivered
24456	2026-10-02 12:14:27.227	Racing Socks, L	4	8.99	35.96	Delivered
9463	2026-09-23 12:14:27.227	Men's Bib-Shorts, S	8	89.99	719.92	Shipped
69914	2026-09-22 12:14:27.227	Front Brakes	10	106.50	1065.00	Pending
30858	2026-09-15 12:14:27.227	ML Road Tire	9	24.99	224.91	Shipped
41047	2026-09-15 12:14:27.227	Road-650 Red, 58	2	782.99	1565.98	Pending
20941	2026-09-12 12:14:27.227	Road-250 Red, 44	8	2443.35	19546.80	Delivered

OrderID	OrderDate	ProductName	Quantity	UnitPrice	TotalAmount	Status
15096	2026-10-07 12:14:27.227	HL Road Frame - Black, 44	7	1431.50	10020.50	Delivered
38966	2026-10-03 12:14:27.227	Touring-1000 Yellow, 60	1	2384.07	2384.07	Delivered
24456	2026-10-02 12:14:27.227	Racing Socks, L	4	8.99	35.96	Delivered
9463	2026-09-23 12:14:27.227	Men's Bib-Shorts, S	8	89.99	719.92	Shipped

Consulta ejecutada correctamente. server-lab6-alumno09.databa... adminlab6 (132) AdventureWorkLT | 0000:01 | Fila: 1, Columna: 1 51.914 filas

Query Store GUI

Archivo Editar Ver Git Proyecto Herramientas Extensiones Ventana Ayuda

Buscar

AdventureWorkLT Ejecutar

Explorador de objetos

server-lab6-alumno09.databa... adminlab6 (132) AdventureWorkLT

Bases de datos

Bases de datos del sistema

AdventureWorkLT

Diagramas de base de datos

Tablas

Vistas

Recursos externos

Sinónimos

Programación

Almacén de consultas

Consultas con regresión

Consumo general de recursos

Consultas que más recursos consumen

Consultas con planes forzados

Consultas con gran variación

Estadísticas de espera de consulta

Consultas con seguimiento

Eventos extendidos

Generador de eventos XEvent

Almacenamiento

Seguridad

Catálogos de Integration Services

Principales 25 consumidores de recursos de la base de datos AdventureWorkLT. Periodo de tiempo: Última hora; fecha de finalización: 07/10/2026 19:40

Métrica Duración (ms) Estadística Total

total duración

id. de consulta

Resumen del plan de la consulta 3

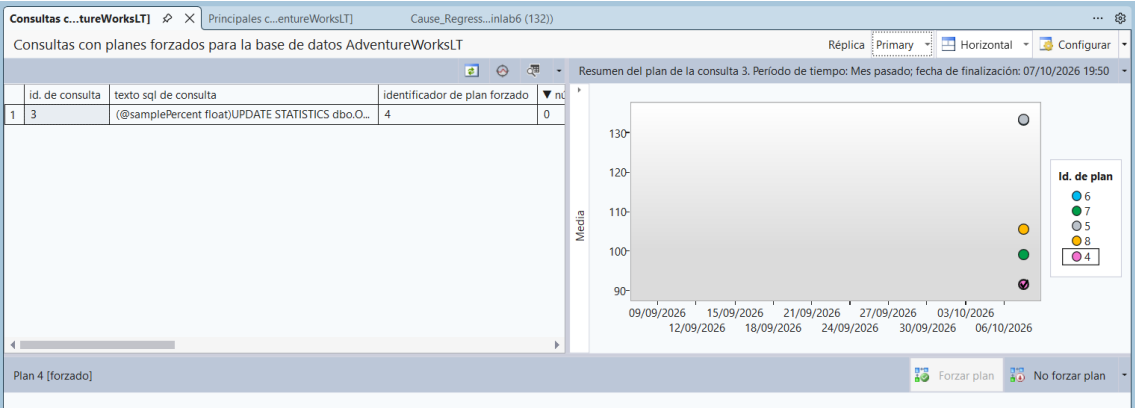
Id. de plan

Plan 5 [no forzado]

Forzar plan No forzar plan

19:45 07/10/2026

Forzar Plan



Verify plan forcing

SQLQuery3.s...lab6 (117))*

```
1  SELECT
2      q.query_id,
3      p.plan_id,
4      p.is_forced_plan,
5      p.force_failure_count,
6      qt.query_sql_text
7  FROM sys.query_store_plan AS p
8  INNER JOIN sys.query_store_query AS q
9      ON p.query_id = q.query_id
10 INNER JOIN sys.query_store_query_text AS qt
11     ON q.query_text_id = qt.query_text_id
12 WHERE p.is_forced_plan = 1;
13 GO
14 EXEC dbo.GetCustomerOrders @CustomerID = 29485;
15 GO
16 EXEC dbo.GetCustomerOrders @CustomerID = 1;
17 GO
18 DROP PROCEDURE IF EXISTS dbo.GetCustomerOrders;
19 GO
20 DROP INDEX IX_OrderHistory_CustomerDate ON dbo.OrderHistory;
21 GO
22 CREATE NONCLUSTERED INDEX IX_OrderHistory_CustomerDate
23 ON dbo.OrderHistory (CustomerID, OrderDate DESC)
24 INCLUDE (ProductID, Quantity, UnitPrice, Status);
```

110 % 3 0

Resultados Mensajes

	query_id	plan_id	is_forced_plan	force_failure_count	query_sql_text
1	3	4	1	0	(@samplePercent float)UPDATE STATISTICS <u>dbo.Order...</u>

	OrderID	OrderDate	ProductName	Quantity	UnitPrice	TotalAmount	Status
10	62212	2026-08-25 12:14:27.227	LL Mountain Frame - Silv...	10	264.05	2640.50	Delivered
11	69026	2026-08-24 12:14:27.227	Women's Mountain Short...	2	69.99	139.98	Pending
12	1529	2026-08-23 12:14:27.227	HL Road Seat/Saddle	1	52.64	52.64	Shipped
13	62883	2026-08-22 12:14:27.227	Rear Brakes	2	106.50	213.00	Delivered
14	69582	2026-08-20 12:14:27.227	Mountain-100 Black, 42	5	3374.99	16874.95	Pending
15	13028	2026-08-17 12:14:27.227	LL Mountain Frame - Silv...	5	277.50	1387.50	Delivered

Consulta ejecutada correctamente.

Apply a Query Store hint

Partitions Query

SQLQuery4.s...lab6 (128)*

```
1 SELECT
2     oh.CustomerID,
3     p.Name AS ProductName,
4     pc.Name AS CategoryName,
5     oh.Status,
6     COUNT(*) AS OrderCount,
7     SUM(oh.TotalAmount) AS TotalRevenue,
8     AVG(oh.TotalAmount) AS AvgOrderValue,
9     STDEV(oh.TotalAmount) AS StdDevOrderValue,
10    RANK() OVER (PARTITION BY oh.Status
11                ORDER BY SUM(oh.TotalAmount) DESC) AS StatusRevenueRank,
12    PERCENT_RANK() OVER (PARTITION BY pc.Name
13                        ORDER BY AVG(oh.TotalAmount)) AS CategoryPctRank,
14    SUM(COUNT(*)) OVER (PARTITION BY oh.Status) AS StatusTotalOrders
15 FROM dbo.OrderHistory AS oh
16 INNER JOIN SalesLT.Product AS p
17     ON oh.ProductID = p.ProductID
18 INNER JOIN SalesLT.ProductCategory AS pc
19     ON p.ProductCategoryID = pc.ProductCategoryID
20 GROUP BY oh.CustomerID, p.Name, pc.Name, oh.Status
21 ORDER BY TotalRevenue DESC;
```

110 % No se encontraron problemas.

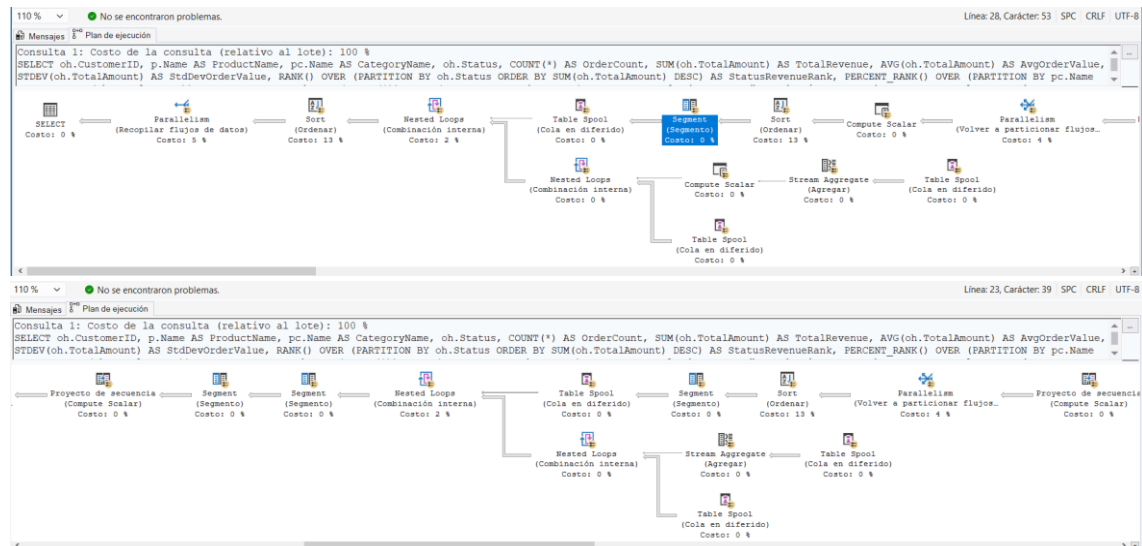
Resultados Mensajes

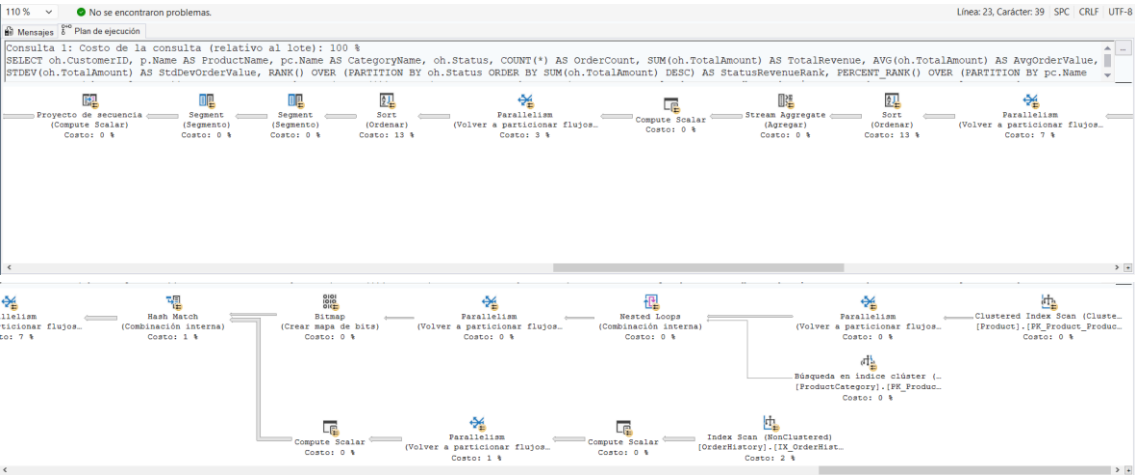
	CustomerID	ProductName	CategoryName	Status	OrderCount	TotalRevenue	AvgOrderValue	StdDevOrderValue	StatusRevenueRank	CategoryPctRank	StatusTotalOrders
1	1	Road-150 Red, 44	Road Bikes	Pending	193	3925362.19	20338.664196	9953.84808121674	1	0.906430268069223	70033
2	1	Road-150 Red, 56	Road Bikes	Pending	163	3435139.20	21074.473619	10237.3929021903	2	0.906515100101798	70033
3	1	Road-150 Red, 62	Road Bikes	Pending	169	3424404.39	20262.747869	10460.7225245103	3	0.906345436036647	70033
4	1	Mountain-100 Silver, 44	Mountain Bikes	Pending	165	3229990.50	19575.700000	10075.8722583865	4	0.828672931469173	70033
5	1	Mountain-100 Silver, 38	Mountain Bikes	Pending	170	3229990.50	18999.944117	9768.9718706274	4	0.828443731377492	70033
6	1	Mountain-100 Silver, 42	Mountain Bikes	Pending	171	3199390.59	18709.886491	9368.58496711959	6	0.828214531285812	70033
7	1	Mountain-100 Black, 44	Mountain Bikes	Pending	166	3121865.75	18806.420180	9663.18807133956	7	0.828329131331653	70033
8	1	Road-150 Red, 48	Road Bikes	Pending	162	3120251.44	19260.811358	10082.5814772644	8	0.887682388870037	70033
9	1	Road-150 Red, 52	Road Bikes	Pending	163	3055842.58	18747.500490	10220.6120669375	9	0.887597556837462	70033
10	1	Mountain-100 Black, 38	Mountain Bikes	Pending	165	2996991.12	18163.582545	9571.14694795577	10	0.808732523493009	70033
11	1	Mountain-100 Black, 48	Mountain Bikes	Pending	155	2959866.23	19095.911161	10054.8805235333	11	0.828558331423333	70033
12	1	Mountain-100 Black, 42	Mountain Bikes	Pending	163	2915001.36	17800.617646	9673.76860016637	12	0.809617031447360	70033

Consulta ejecutada correctamente.

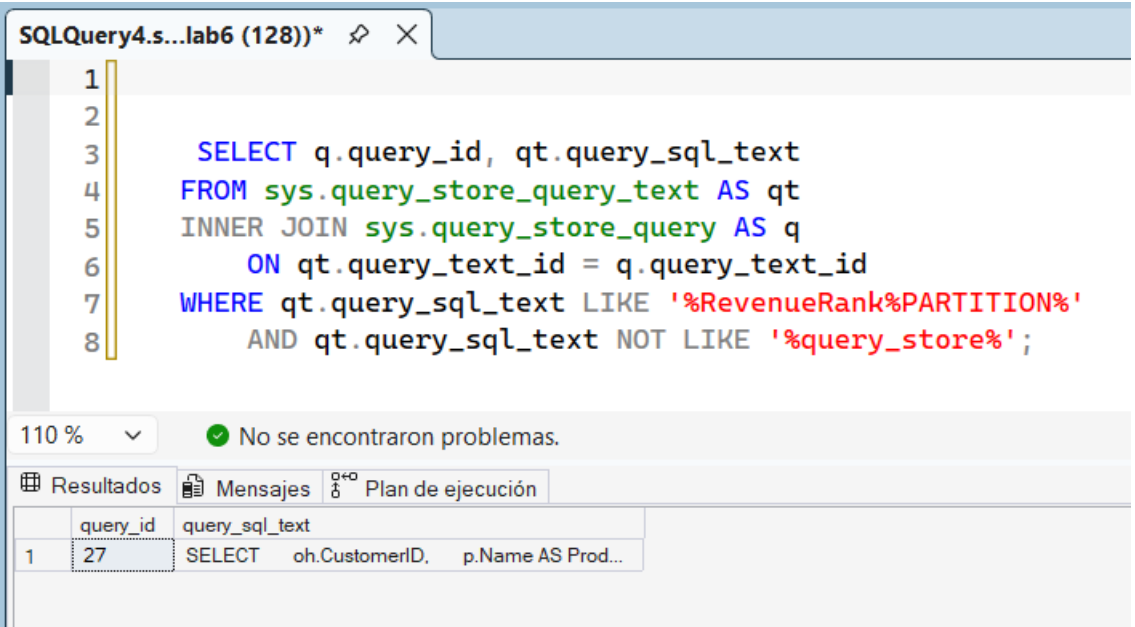
server-lab6-alumno09.databa... adminlab6 (128) AdventureWorksLT 0

Query Store Hint Plan





Find Query_id



Aply MAXDROP

```
3 EXEC sp_query_store_set_hints
4   @query_id = 27,
5   @query_hints = N'OPTION (MAXDOP 1)';
```

0 % ▾  No se encontraron problemas.

Mensajes

Los comandos se han completado correctamente.

Hora de finalización: 2026-10-07T20:07:05.8985214+02:00

Clear Hint

```
2
3 EXEC sp_query_store_clear_hints @query_id = 27;
```

110 % ▾  No se encontraron problemas.

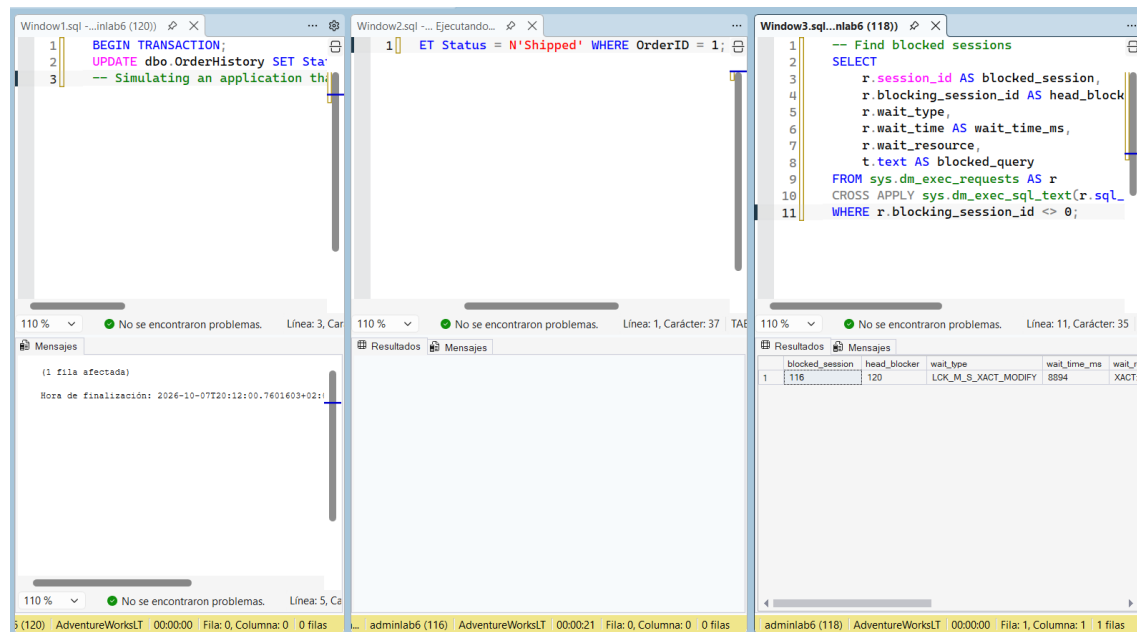
Mensajes

Los comandos se han completado correctamente.

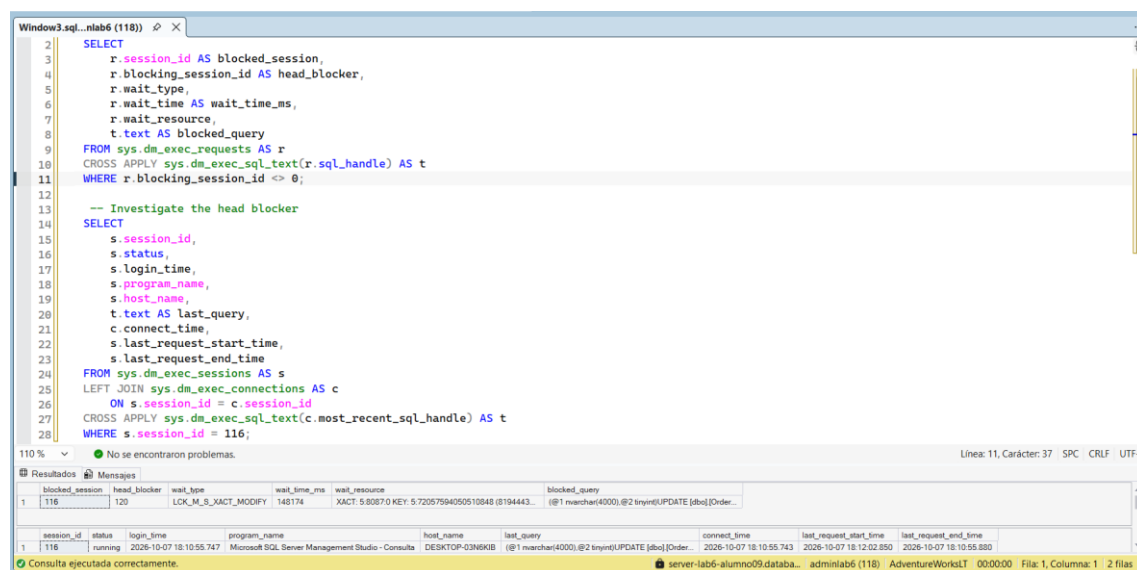
Hora de finalización: 2026-10-07T20:09:11.0703930+02:00

Identify and resolve blocking

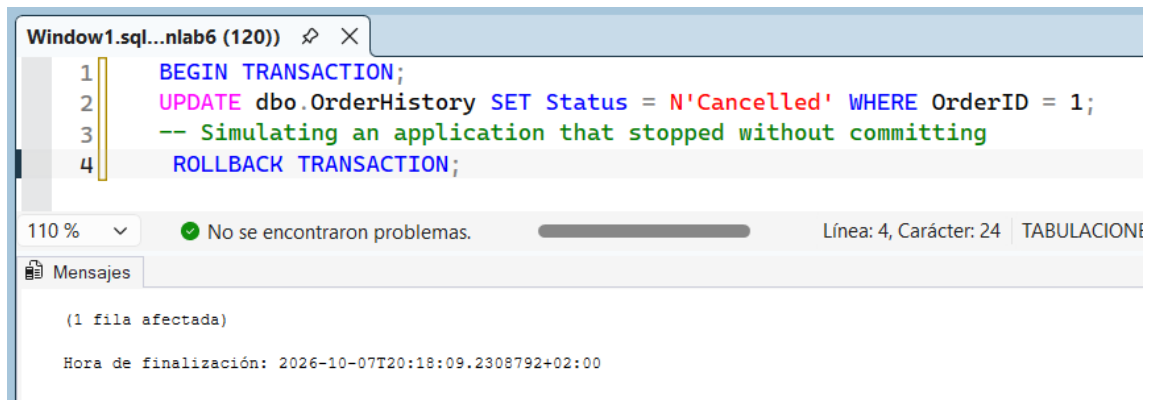
3 Windows



Investigate Head Blocker



Window 1 Rollback



The screenshot shows a SQL Server Enterprise Manager window titled 'Window1.sql...nlab6 (120)'. The SQL script contains four lines: 1. BEGIN TRANSACTION; 2. UPDATE dbo.OrderHistory SET Status = N'Cancelled' WHERE OrderID = 1; 3. -- Simulating an application that stopped without committing 4. ROLLBACK TRANSACTION; The status bar indicates '110 %', 'No se encontraron problemas.', 'Línea: 4, Carácter: 24', and 'TABULACIONES'. The 'Mensajes' pane shows '(1 fila afectada)' and 'Hora de finalización: 2026-10-07T20:18:09.2308792+02:00'.

```
1 BEGIN TRANSACTION;
2 UPDATE dbo.OrderHistory SET Status = N'Cancelled' WHERE OrderID = 1;
3 -- Simulating an application that stopped without committing
4 ROLLBACK TRANSACTION;
```

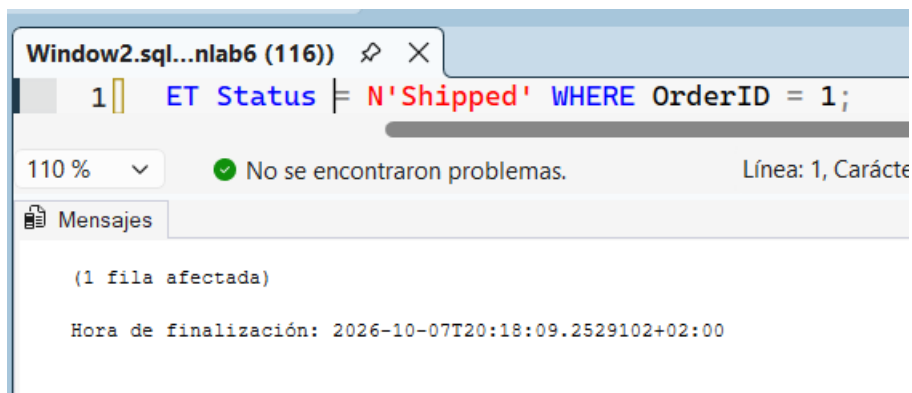
110 % No se encontraron problemas. Línea: 4, Carácter: 24 TABULACIONES

Mensajes

(1 fila afectada)

Hora de finalización: 2026-10-07T20:18:09.2308792+02:00

Window 2 Executed



The screenshot shows a SQL Server Enterprise Manager window titled 'Window2.sql...nlab6 (116)'. The SQL script contains one line: 1. SET Status = N'Shipped' WHERE OrderID = 1; The status bar indicates '110 %', 'No se encontraron problemas.', and 'Línea: 1, Carácter: 24'. The 'Mensajes' pane shows '(1 fila afectada)' and 'Hora de finalización: 2026-10-07T20:18:09.2529102+02:00'.

```
1 SET Status = N'Shipped' WHERE OrderID = 1;
```

110 % No se encontraron problemas. Línea: 1, Carácter: 24

Mensajes

(1 fila afectada)

Hora de finalización: 2026-10-07T20:18:09.2529102+02:00